Problem Set 2: Volatility

MGT 927

We use daily CRSP data from WRDS for 2 January 1990 to 31 December 2020, which gives 7,812 trading days. The market series is the CRSP value-weighted return (`vwretd`) and MSFT is its CRSP holding-period return, both in percent per day. We code every likelihood by hand and maximize it numerically. Standard errors are the Bollerslev–Wooldridge robust sandwich, and the point estimates agree with the `arch` Python package to three decimals. The code is in `volatility_ps.py`.

	Mean	S.d.	Ann. vol	Skew	Excess kurt.	Min	Max
Market	0.044	1.13	17.9%	−0.32	11.0	−11.8	11.5
MSFT	0.102	2.02	32.1%	0.23	6.5	−15.6	19.6

Question 1

(a) Autocorrelations of returns and squared returns

Before looking at the data we expected the market to show the stronger volatility clustering. Write MSFT's return as $r_{i,t} = \beta r_{m,t} + e_{i,t}$. The part of volatility that persists for weeks or months comes from economy-wide conditions such as recessions, financial stress and uncertainty about policy, and the index is exposed to all of it. MSFT carries the same market component plus firm-specific news: earnings announcements, product news and, in this sample, an antitrust case. That kind of news tends to arrive as a single large move that is not followed by more large moves. Each such day adds a big value of $r_{i,t}^2$ that has no counterpart in the days around it, so it raises the variance of squared returns without raising their autocovariance, and the measured autocorrelation falls. Diversification strips this component out of the index. For returns themselves we expected autocorrelations close to zero in both series, since predictable daily returns in liquid stocks would be traded away.

The data confirm the prior. Squared demeaned returns are positively autocorrelated at all 20 lags in both series, so both show volatility clustering, but the market's clustering is much stronger. Its autocorrelations average 0.256 over the 20 lags against 0.103 for MSFT, and its Ljung–Box $Q(20)$ is 11,056 against 1,853. The market's autocorrelations also fade more slowly, keeping 57% of their lag-1 value at lag 20 (0.176 against 0.311), while MSFT's keep 26% (0.048 against 0.182). The same ordering holds for absolute returns, which are much less sensitive to a few extreme days, so the result does not come from outliers.

Two further facts support the mechanism behind the prior. First, the firm-specific part of MSFT's return, the residual from a regression on the market, has squared-return autocorrelations that average only 0.062. MSFT's autocorrelations sit between that value and the market's, which is what one would expect if they are a blend of a strongly clustered market component and a weakly

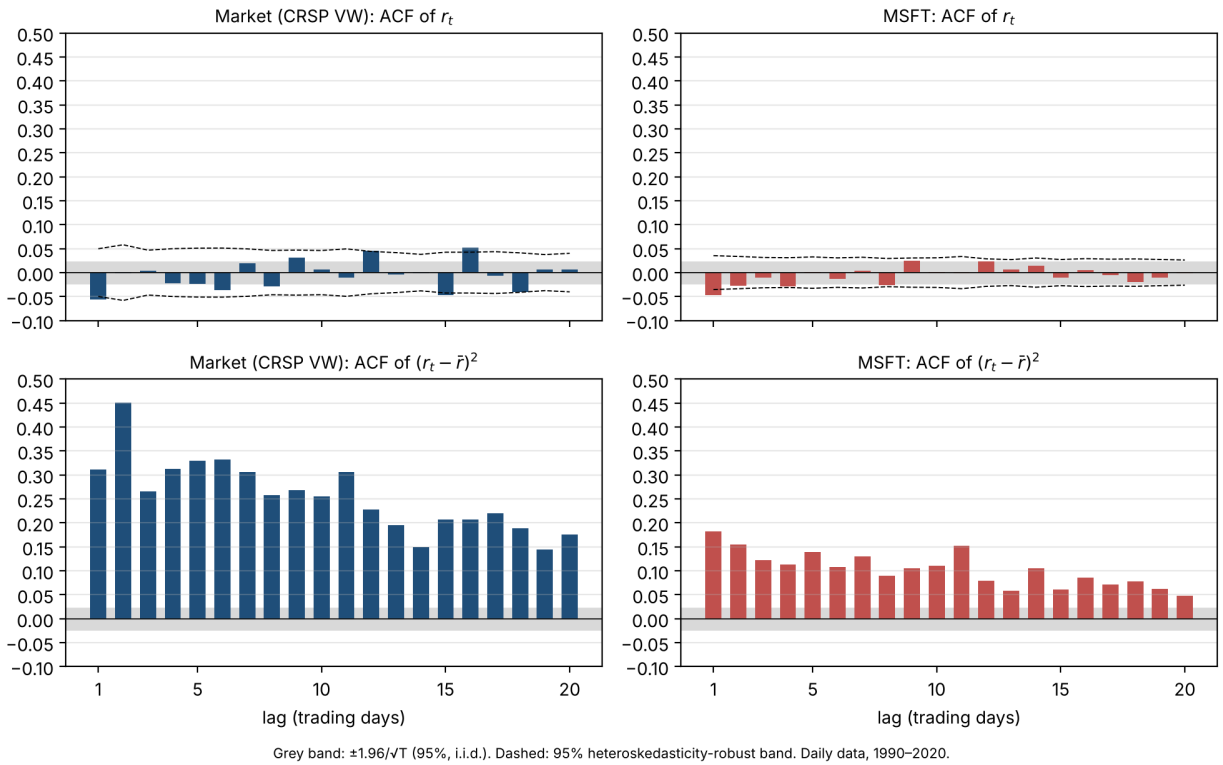


Figure 1: Autocorrelations at lags 1–20. Grey band: i.i.d. 95% band $\pm 1.96/\sqrt{T}$. Dashed: heteroskedasticity-robust 95% band.

clustered firm component. Second, three of MSFT's five largest daily moves were firm-specific jumps: +19.6% on 19 October 2000, −15.6% on 24 April 2000 and −14.5% on 3 April 2000, days on which the market moved by less than 4%. The other two, +18.6% on 13 October 2008 and −14.7% on 16 March 2020, came on days when the market itself moved by more than 11%. The unconditional excess kurtosis is higher for the market (11.0 against 6.5), but that is because volatility clustering itself fattens the unconditional tails. Once the GARCH model in part (b) removes the clustering, the excess kurtosis of the standardized residuals is 5.8 for MSFT against 1.9 for the market, which is the signature of jumps that the volatility state does not predict.

The return autocorrelations are small, as expected. None exceeds 0.06 in absolute value. Against the usual $\pm 1.96/\sqrt{T}$ band, 10 of the market's 20 lags and 6 of MSFT's look significant, but that band assumes constant variance and is too narrow here. With a band that allows for time-varying variance, 4 market lags and 1 MSFT lag remain significant, and the largest coefficient, −0.056 for the market at lag 1, implies that less than 0.4% of the return variance is predictable from the previous day.

(b) Gaussian GARCH(1,1) by maximum likelihood

The model is

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \quad z_t \overset{iid}{\sim} N(0,1), \qquad \sigma_t^2 = \omega + \alpha\, \varepsilon_{t-1}^2 + \beta\, \sigma_{t-1}^2,$$

with $\omega > 0$, $\alpha, \beta \geq 0$ and $\alpha + \beta < 1$. Conditional on $\mathcal{F}_{t-1}$, $r_t \sim N(\mu, \sigma_t^2)$ and σ_t^2 is a known function of $\theta = (\mu, \omega, \alpha, \beta)$ and past returns. The log likelihood is therefore the sum of the one-step conditional normal log densities,

$$\mathcal{L}(\theta) = -\frac{1}{2}\sum_{t=1}^{T}\left[\log(2\pi) + \log\sigma_t^2(\theta) + \frac{(r_t-\mu)^2}{\sigma_t^2(\theta)}\right], \qquad \hat{\theta} = \arg\max_{\theta}\mathcal{L}(\theta),$$

where the variance recursion starts from $\sigma_1^2 = T^{-1}\sum_t (r_t - \mu)^2$.

	μ	ω	α	β	$\alpha+\beta$	Half-life	Long-run vol	$\log L$
Market	0.071	0.0175	0.112	0.873	0.985	47	17.4%	−10,110.2
	(0.009)	(0.0035)	(0.012)	(0.012)	(0.005)			
MSFT	0.116	0.066	0.079	0.906	0.985	47	33.9%	−15,649.6
	(0.018)	(0.036)	(0.025)	(0.030)	(0.009)			

Table 1: GARCH(1,1) estimates; robust s.e. in parentheses. The s.e. of $\alpha+\beta$ uses the delta method with the full covariance matrix. Half-life (days) $= \log 0.5/\log(\alpha+\beta)$ for deviations of the variance from its long-run level. Long-run vol $= \sqrt{252\,\omega/(1-\alpha-\beta)}$.

Both fits have the usual daily-data shape. A return surprise moves next-day variance only a little (α around 0.1), but the variance remembers its past almost completely (β around 0.9), so $\alpha+\beta = 0.985$ and a deviation of the variance from its long-run level takes about 47 trading days to shrink by half. Whether persistence is significantly below one depends on the standard error of the sum, and that standard error cannot be read off the two individual ones. The estimates of α and β are strongly negatively correlated (-0.93 for the market and -0.97 for MSFT), because the likelihood can trade a higher α against a lower β with little change in fit. With the covariance included, the standard error of $\alpha + \beta$ is 0.0046 for the market and 0.0086 for MSFT. A one-sided test of the integrated null $\alpha + \beta = 1$ rejects clearly for the market ($t = 3.2$) and only marginally for MSFT ($t = 1.7$, $p \approx 0.045$), so MSFT's long-run variance is estimated much less precisely.

The two series differ in how they reach the same persistence. The market's variance reacts more to each surprise ($\alpha = 0.112$ against 0.079), which is the model's version of its higher squared-return autocorrelations in part (a), while MSFT relies more on its own lagged variance. The implied long-run volatilities of 17.4% and 33.9% are close to the sample volatilities of 17.9% and 32.1%. On the standardized residuals, $Q(20)$ of the squares falls to 25.2 ($p = 0.19$) for the market and 11.6 ($p = 0.93$) for MSFT, so one GARCH(1,1) equation captures essentially all of the clustering. It does not capture the tails: the standardized residuals still have excess kurtosis of 1.9 and 5.8, against zero under normality. The Gaussian likelihood is therefore a quasi-likelihood. The estimates remain consistent if the mean and variance equations are right, but the classical standard errors are not valid. The difference is concentrated in the variance parameters: for MSFT's ω, α and β the robust standard errors are 2.7 to 3.0 times the classical ones, while for μ the two are almost identical.

(c) Is MSFT's volatility just the market moving?

If the risk manager were right, MSFT's conditional variance would be close to $b^2\sigma_{m,t}^2$, and three things would follow. The market would explain most of MSFT's return variance, the firm-specific residual would carry no volatility dynamics of its own, and MSFT's variance would rise and fall one-for-one with the market's. The estimates reject all three.

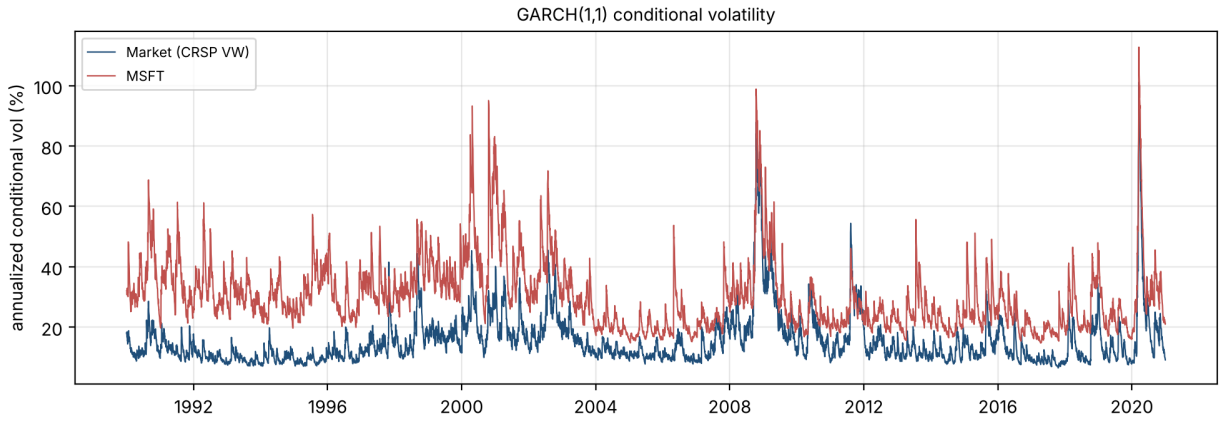


Figure 2: GARCH(1,1) conditional volatility, annualized.

The market model $r_{MSFT,t} = a + b\, r_{m,t} + u_t$ gives $\hat{b} = 1.15$ (White s.e. 0.021) and $R^2 = 0.41$. Over 1990–2020 the market accounts for 41% of MSFT's daily variance, and the other 59% is firm-specific. The residual is also far from noise. A GARCH(1,1) fitted to u_t gives $\alpha = 0.024$ (0.006) and $\beta = 0.971$ (0.007), with persistence 0.995 (0.002), and $Q(20)$ of u_t^2 is 704. Firm-specific volatility therefore has its own slow-moving state, at least as persistent as the market's, and a beta-scaled market forecast would miss it entirely. Finally, a regression of MSFT's fitted log variance on the market's fitted log variance has a slope of 0.50 and an R^2 of 0.39. Under the claim both would be close to one. A doubling of market variance is associated with roughly a 40% rise in MSFT's variance, and most of the movement in MSFT's variance is unrelated to the market's.

The two GARCH fits also show when the claim comes closest to holding. Combining them gives an approximate market-related share of MSFT's variance,

$$s_t = \frac{b^2 \hat{\sigma}^2_{m,t}}{b^2 \hat{\sigma}^2_{m,t} + \hat{\sigma}^2_{u,t}}.$$

This is an approximation, not an exact conditional decomposition, because it assumes a constant beta and a market return that is conditionally uncorrelated with u_t. OLS only guarantees zero correlation on average, and yearly betas range from 0.73 to 2.30. As a check, the implied total variance averages 1.05 times MSFT's own GARCH variance, with a 10th–90th percentile range of 0.72 to 1.43. The share averages 0.36 and exceeds one half on only 20% of days.

Period	Approx. market share	Market-related vol	Firm-specific vol
Full sample 1990–2020	0.36	17.8%	23.5%
Dot-com and antitrust 2000–02	0.36	25.3%	33.8%
Financial crisis Sep 2008–Jun 2009	0.64	49.4%	33.7%
COVID crash Mar–Apr 2020	0.87	71.3%	24.2%
Calm year 2017	0.32	10.0%	14.7%

Table 2: Average approximate market-related share of MSFT's conditional variance, and the average annualized volatility of each part.

The share is high only in market-wide crises. In the 2008–09 financial crisis it averaged 0.64 and in March–April 2020 it reached 0.87, because market volatility exploded while MSFT's firm-specific

volatility rose much less. In those months the risk manager's description is roughly right. In calm markets it is wrong (0.32 in 2017), and in 2000–02 it is wrong for the opposite reason: MSFT's volatility was very high, but the antitrust case and the end of the technology boom raised its firm-specific volatility (33.8%) as much as its market-related volatility, so the share stayed at its average. The relationship has tightened in recent years, with a yearly R^2 of 0.64 to 0.75 in 2018–2020, as Microsoft became one of the largest weights in the index. Taken over the whole sample, though, most of MSFT's risk is its own, and a risk model that treats MSFT as leveraged market exposure would understate its risk in normal times and miss firm-specific episodes altogether.

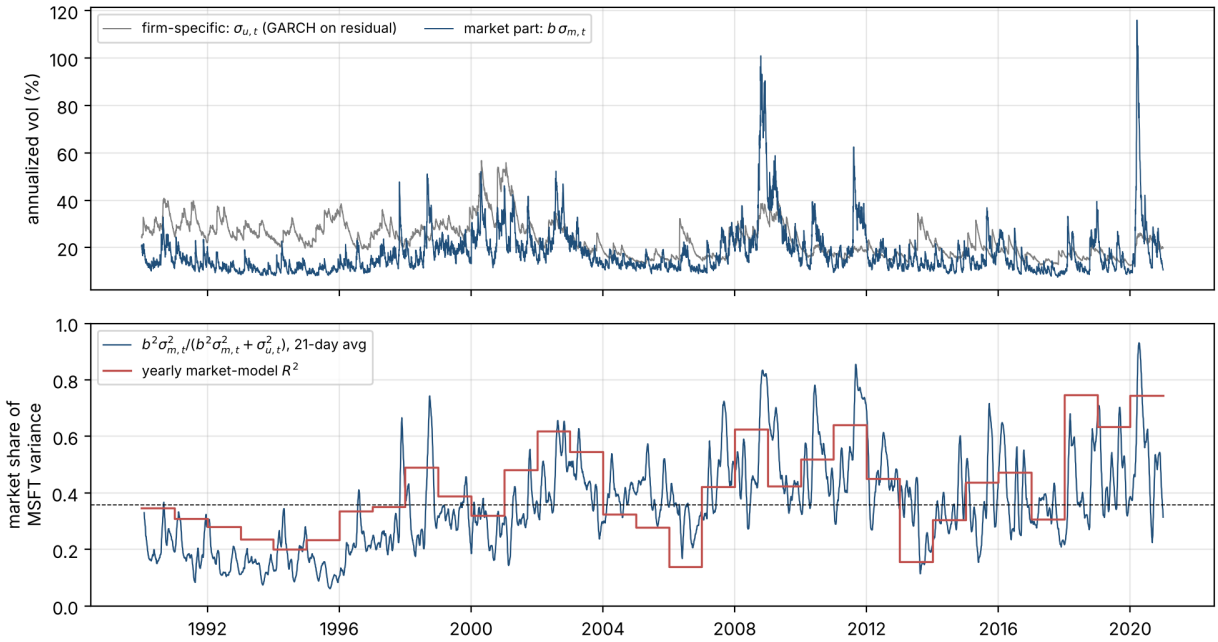


Figure 3: Top: market-related ($b\,\sigma_{m,t}$) and firm-specific ($\sigma_{u,t}$) volatility of MSFT. Bottom: approximate market-related share of MSFT's variance (dashed: mean) and yearly market-model R^2.

(d) Market risk over the next month (as of 31 December 2020)

The forecasts come from the fitted GARCH(1,1) for the market. The variance forecast h days ahead is $\mathbb{E}_t\sigma^2_{t+h} = \bar{\sigma}^2 + (\alpha+\beta)^{h-1}(\sigma^2_{t+1} - \bar{\sigma}^2)$, so a gap between today's variance and its long-run level halves in about 47 trading days. The one-month variance is the sum of the 21 daily forecasts, which gives a one-month standard deviation of 3.0% against 5.0% at the long-run level. The ranges and loss probabilities below come from 200,000 simulated 21-day paths of the fitted model, because a month of GARCH returns is not exactly normal. They set the expected return to zero; the fitted mean of 0.071% a day, about 18% a year, would shift every figure up by 1.5 percentage points. For a reader who has never heard of GARCH, the model says three things.

- Risk is low right now. Recent daily moves have been small, so the model puts the market's volatility at about 9% a year, half its long-run level of 17%. Calm periods tend to last.
- It will rise only slowly. Volatility drifts back toward normal over several months and reaches only about 12% by the end of January. A typical month should move the market about 3% up or down, against 5% normally, and 19 months in 20 like this one end between −6% and

+6%.

- A bad month is still possible. About 1 month in 100 like this one loses more than 7.5%, and one large shock would raise all of these numbers at once: in March 2020 the same model put volatility near 100%. Real crashes are also more likely than the model's bell-curve shocks allow.

Question 2 (PhD): Gaussian GARCH-in-mean

In the GARCH-in-mean model the conditional variance enters the expected return, $r_t = \mu + \lambda\, g(\sigma_t^2) + \varepsilon_t$ with $\varepsilon_t = \sigma_t z_t$, $z_t \sim N(0,1)$ and $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, where $g(\sigma^2)$ is usually σ^2, σ or $\log\sigma^2$. The MLE solves

$$\hat{\theta} = \arg\max_{\theta} \; -\frac{1}{2}\sum_{t=1}^{T}\left[\log 2\pi + \log\sigma_t^2(\theta) + \frac{\left(r_t - \mu - \lambda g(\sigma_t^2(\theta))\right)^2}{\sigma_t^2(\theta)}\right] \quad \text{s.t. } \omega > 0,\ \alpha, \beta \geq 0,\ \alpha+\beta < 1.$$

The difference from plain GARCH is that the residual $\varepsilon_t = r_t - \mu - \lambda g(\sigma_t^2)$ now depends on the variance parameters, so the mean and variance recursions have to be run together: σ_t^2 is built from ε_{t-1}, and ε_t is then built from σ_t^2. For the same reason the information matrix is no longer block-diagonal between mean and variance parameters. A misspecified variance equation now biases the estimate of λ, and the two blocks cannot be estimated in separate steps. Because a misspecified variance equation biases λ, the estimate should come from a variance equation the data support. The table below estimates the model with $g(\sigma^2) = \sigma^2$ under each of the variance equations compared in Question 3.

Variance equation	Market $\hat{\lambda}$	MSFT $\hat{\lambda}$	CAPM benchmark for MSFT
GARCH-N	0.041 (0.013)	0.012 (0.009)	0.015
GARCH-t	0.034 (0.012)	0.005 (0.007)	0.012
GJR-N	0.021 (0.011)	0.008 (0.009)	0.008
GJR-t	0.023 (0.010)	0.004 (0.007)†	0.008

Table 3: GARCH-in-mean estimates of λ with $g(\sigma^2) = \sigma^2$; robust s.e. in parentheses. The CAPM benchmark is $\lambda_m R^2/b$ with the market $\hat{\lambda}$ from the same row. †MSFT's variance persistence is on the imposed bound, so this s.e. is only indicative.

For the market, adding the leverage term roughly halves $\hat{\lambda}$, from 0.041 under the symmetric Gaussian GARCH to 0.023 (0.010) under GJR-t, the specification Question 3 favors. Which days count as high-variance depends on the variance equation, and λ is estimated from the returns on those days, so the bias described above is large in these data. The risk–return trade-off stays positive and is marginally significant under GJR-t ($t = 2.4$), so the conclusion survives, but it is about half as strong as the symmetric model suggests. This is the problem the GJR model was introduced to address; in its first application, allowing for asymmetric volatility turned the estimated risk–return relation negative.

For MSFT, $\hat{\lambda}$ is small and insignificant under every variance equation, and each estimate is within one standard error of what CAPM implies. If only market risk is priced, MSFT's premium is $b\,\lambda_m\sigma_{m,t}^2$, so its price per unit of total variance should be $\lambda_m R^2/b$: 0.015 under the Gaussian GARCH and 0.008 under GJR-t, against estimates of 0.012 and 0.004. MSFT's λ is lower than the market's because most of its variance is firm-specific and earns no premium.

Additional return predictors enter the mean equation. With $m_t = \mu + \lambda g(\sigma_t^2) + \gamma' X_{t-1}$ the objective is unchanged except that $\varepsilon_t = r_t - m_t$, and the variance recursion is fed this new residual. The predictors must be known at $t-1$, and they must not be collinear with $g(\sigma_t^2)$. If one of them is itself a volatility proxy, such as a lagged VIX, λ and γ are only weakly identified.

Variance predictors enter the GARCH equation, $\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 + \delta' Q_{t-1}$, again with the same objective, but the restrictions that make plain GARCH well behaved no longer cover the new term. The first problem is positivity. The conditions $\omega > 0$ and $\alpha, \beta \geq 0$ guarantee $\sigma_t^2 > 0$ in plain GARCH, but $\delta' Q_{t-1}$ can be negative, and a single negative variance makes the likelihood undefined. The simplest fix is to use non-negative predictors, such as variances or squared spreads, together with $\delta \geq 0$. For predictors of either sign one has to restrict $\omega + \delta' Q_t > 0$ over the support of Q or reparametrize, for example with $\exp(\delta' Q_{t-1})$ or an EGARCH-X model for $\log \sigma_t^2$. The predictors must also be predetermined, $Q_{t-1} \in \mathcal{F}_{t-1}$; otherwise σ_t^2 is no longer a conditional variance and the prediction-error decomposition behind the likelihood breaks down.

The second problem is stationarity. With an exogenous input, $\alpha + \beta < 1$ is no longer enough, because the variance inherits the behaviour of Q. Q_t must be strictly stationary and ergodic with $\mathbb{E}|Q_t| < \infty$, in which case $\mathbb{E}\sigma^2 = (\omega + \delta' \mathbb{E}Q)/(1 - \alpha - \beta)$; a trending or unit-root predictor would make σ_t^2 non-stationary however small $\alpha + \beta$ is. These conditions, together with identification, give consistency of the (Q)MLE. Asymptotic normality needs more: an interior true parameter, $\mathbb{E}z_t^4 < \infty$ so that the score has finite variance, and enough moments of Q_t and σ_t^2 for the score and Hessian to obey a law of large numbers and a central limit theorem. Identification requires that Q vary over time and not be collinear with ε_{t-1}^2 or σ_{t-1}^2, since otherwise ω, δ and β cannot be separated. Finally, the sign restriction matters for testing. If $\delta \geq 0$ is imposed, $\delta = 0$ lies on the boundary of the parameter space, and for a scalar δ with the other parameters interior the likelihood-ratio statistic converges to $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ rather than χ_1^2.

Question 3 (PhD): GJR-GARCH(1,1,1) with Student-t innovations

The model is $r_t = \mu + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$, with z_t i.i.d. standardized Student-t (mean 0, variance 1, $\nu > 2$) and

$$\sigma_t^2 = \omega + (\alpha + \gamma\, \mathbf{1}\{\varepsilon_{t-1} < 0\})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2,$$

so a negative return raises next-day variance by $\alpha + \gamma$ per unit of squared return and a positive return by α. The log likelihood is

$$\mathcal{L} = \sum_t \left[\log\Gamma\!\left(\tfrac{\nu+1}{2}\right) - \log\Gamma\!\left(\tfrac{\nu}{2}\right) - \tfrac{1}{2}\log\left(\pi(\nu-2)\right) - \tfrac{1}{2}\log\sigma_t^2 - \tfrac{\nu+1}{2}\log\!\left(1 + \frac{\varepsilon_t^2}{(\nu-2)\sigma_t^2}\right)\right].$$

To derive the non-integration condition, substitute $\varepsilon_{t-1}^2 = \sigma_{t-1}^2 z_{t-1}^2$ and note that $\mathbf{1}\{\varepsilon_{t-1} < 0\} = \mathbf{1}\{z_{t-1} < 0\}$ because $\sigma_{t-1} > 0$:

$$\sigma_t^2 = \omega + \sigma_{t-1}^2\big[\alpha z_{t-1}^2 + \gamma z_{t-1}^2 \mathbf{1}\{z_{t-1} < 0\} + \beta\big].$$

z_{t-1} is independent of $\sigma_{t-1}^2 \in \mathcal{F}_{t-2}$ and $\mathbb{E}z^2 = 1$. Let $\kappa = \mathbb{E}[z^2 \mathbf{1}\{z < 0\}]$ be the part of $\mathbb{E}z^2$ that comes from negative shocks. Taking expectations at $t-2$,

$$\mathbb{E}_{t-2}\sigma_t^2 = \omega + \phi\, \sigma_{t-1}^2, \qquad \phi = \alpha + \kappa\gamma + \beta.$$

The model assumes a standardized t, which is symmetric about zero, so $\kappa = \frac{1}{2}$ and $\phi = \alpha + \frac{1}{2}\gamma + \beta$. Iterating gives $\mathbb{E}_{t-k}\sigma_t^2 = \omega \sum_{j=0}^{k-2} \phi^j + \phi^{k-1}\sigma_{t-k+1}^2$. The effect of today's variance on the forecast k periods ahead decays like ϕ^k, and the unconditional variance $\mathbb{E}\varepsilon_t^2 = \omega/(1-\phi)$ is finite if and only if

$$\boxed{\alpha + \tfrac{1}{2}\gamma + \beta < 1} \qquad (\omega > 0,\ \alpha \geq 0,\ \alpha + \gamma \geq 0,\ \beta \geq 0,\ \nu > 2).$$

At $\phi = 1$ the variance is integrated: a shock never dies out of the variance forecast, and the unconditional variance is infinite. This does not mean volatility explodes. Strict stationarity only requires $\mathbb{E}\log(\alpha z^2 + \gamma z^2 \mathbf{1}\{z < 0\} + \beta) < 0$, and at $\phi = 1$ this holds by Jensen's inequality, since $\mathbb{E}\log X < \log \mathbb{E}X = 0$ for any non-degenerate X. An integrated GJR process can therefore be strictly stationary, and the effect of a shock on future variance still dies out with probability one even though its effect on the expected variance does not.

	μ	ω	α	γ	β	ν	ϕ	$\log L$
Market	0.065	0.0137	0.009	0.163	0.895	6.9	0.985	−9,832.4
	(0.008)	(0.0024)	(0.006)	(0.018)	(0.010)	(0.57)	(0.003)	
MSFT†	0.073	0.032	0.056	0.068	0.910	5.3	0.99999	−15,220.2
	(0.016)	(0.010)	(0.010)	(0.015)	(0.014)	(0.32)	–	

Table 4: GJR-GARCH-t estimates; robust s.e. in parentheses. $\phi = \alpha + \gamma/2 + \beta$ uses the model's symmetry assumption; with the sample weight on negative shocks ($\hat{\kappa} = 0.583$ for the market) the market's ϕ is 0.999. †The constraint $\phi \leq 1 - 10^{-5}$ imposed in estimation binds for MSFT ($\hat{\phi} = 0.99999$). Without it $\hat{\phi} = 1.0017$ with a log-likelihood gain of 0.12 (LR = 0.24). The MSFT standard errors are computed as if the estimate were interior and are only indicative.

	GARCH-N	GARCH-t	GJR-N	GJR-t
Market	−10,110.2	−9,927.3	−9,999.7	−9,832.4
MSFT	−15,649.6	−15,238.3	−15,624.5	−15,220.2

Table 5: Log likelihood by model, showing how much fat tails and leverage each add. All four likelihoods are coded by hand and use the same variance initialization.

Relative to the Gaussian GARCH(1,1), GJR-t raises the log likelihood by 278 for the market and 429 for MSFT, far more than two extra parameters could produce by chance. The second table shows where the gain comes from. Fat tails matter for both series ($\hat{\nu} = 6.9$ and 5.3; switching to t errors alone adds 183 and 411), and they matter most for MSFT, whose firm-specific jumps are exactly the extreme standardized shocks a normal distribution cannot produce. Leverage matters a great deal for the market (it adds 110) and little for MSFT (25).

The parameters move in ways that follow from this. For the market, the symmetric coefficient falls from 0.112 to an insignificant 0.009 and the asymmetry term $\gamma = 0.163$ takes over, so the GARCH(1,1) α was an average of a strong response to bad news and almost no response to good news. Under the model's symmetry assumption the market's persistence is $\phi = 0.985$ (s.e. 0.0035), apparently well below one, with an implied long-run volatility of 15.2%. Both numbers rest on negative shocks carrying half of $\mathbb{E}z^2$, and the data reject that. The market's standardized residuals are negatively skewed (skewness −0.59), and negative shocks carry 58.3% (approximate s.e. 1.2%) of their squared mass. Because the market's γ is large, the difference matters: with the sample weight, $\phi = 0.9987$ (approximate s.e. 0.0041), which is not significantly below one. This standard error is a plug-in calculation that ignores the covariance between the fitted parameters and the estimated

weight. Within one standard error ϕ reaches one, so the implied long-run volatility ranges from about 25% a year to infinity and is no longer pinned down. The GJR-t fit should therefore not be used for the market's long-run volatility level. The GARCH(1,1) estimate of 17.4%, close to the sample's 17.9%, is the better guide, and a skewed-t density would be the natural next step. For MSFT persistence rises to the bound $\phi \leq 1 - 10^{-5}$. Under t errors the largest firm-specific moves are treated as draws from a fat tail, so they no longer pull β down, and the fitted variance becomes smoother and more persistent. Dropping the bound gives $\hat{\phi} = 1.0017$ with a log-likelihood gain of only 0.12, so the data cannot distinguish MSFT's variance in this model from an integrated process. Its long-run variance is infinite at the unconstrained estimate and cannot be estimated reliably, and its standard errors rest on interior asymptotics that do not apply at the boundary. MSFT's residuals are close to symmetric ($\hat{\kappa} = 0.47$), so this conclusion does not depend on the symmetry assumption.

The leverage effect is far stronger for the market. Bad news raises the market's next-day variance 19 times as much as good news of the same size ($\alpha + \gamma = 0.171$ against $\alpha = 0.009$), while for MSFT the ratio is 2.2 (0.124 against 0.056). This is hard to reconcile with the financial-leverage explanation, in which a falling stock price raises the debt-to-equity ratio and therefore equity volatility. Microsoft had no long-term debt until 2009 and little relative to its market value afterwards, yet the asymmetry is weak for MSFT and very strong for the index. A systematic channel fits better. Under volatility feedback, news that raises expected market volatility raises required returns and pushes prices down at the same moment, and correlations across stocks rise in selloffs, so falling prices and rising volatility arrive together in the aggregate. Volatility feedback requires a positive price of risk, and Question 2 finds one under the GJR-t variance equation ($\hat{\lambda} = 0.023$, s.e. 0.010). Firm-specific news works differently, because a large earnings surprise raises uncertainty about the firm whether it is good or bad. In the fitted model, the +19.6% move on 19 October 2000 lifted MSFT's annualized volatility from 47% to 86% the next day. Fitting the same GJR-t model to MSFT's market-model residual makes the point directly: the firm-specific asymmetry is $\gamma = 0.017$ (s.e. 0.023), indistinguishable from zero. MSFT's own γ is 0.42 times the market's, close to the 0.36–0.41 share of its variance that is market-related, so its asymmetry comes almost entirely through its market exposure.

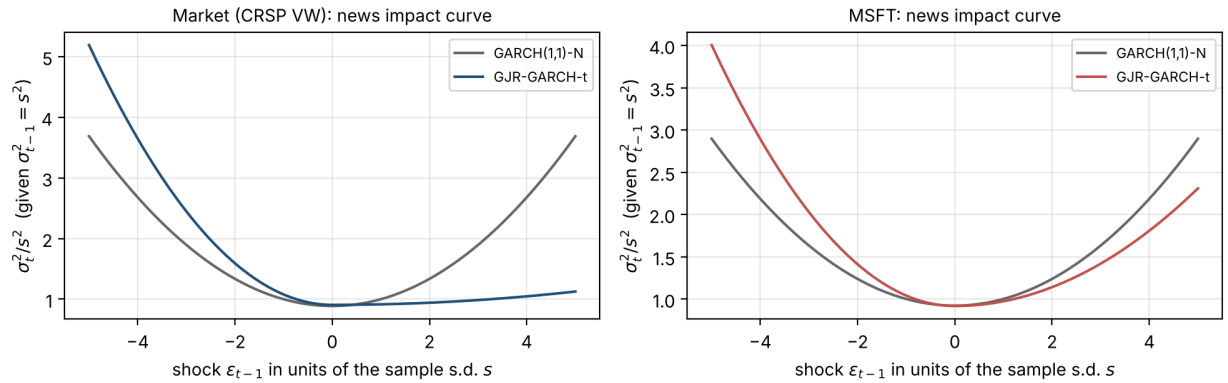


Figure 4: News impact curves: next-day variance after a shock ε_{t-1}, given that the previous variance equals the sample variance s^2. Axes are scaled by the sample s.d. because the long-run variance does not exist for MSFT's GJR-t estimate.